

Table S1. The median UHS at the Dehradun site at eight return periods.

Period (s)	Return Period							
	500 years (g)	1000 years (g)	1500 years (g)	2000 years (g)	2500 years (g)	3500 years (g)	4500 years (g)	5700 years (g)
0.00	0.44	0.59	0.72	0.81	0.88	1.00	1.09	1.18
0.02	0.45	0.61	0.74	0.83	0.90	1.02	1.11	1.20
0.05	0.62	0.80	1.00	1.13	1.23	1.40	1.51	1.63
0.10	0.90	1.17	1.46	1.64	1.79	2.02	2.17	2.32
0.20	1.00	1.43	1.72	1.93	2.07	2.35	2.58	2.81
0.50	0.63	0.92	1.13	1.31	1.43	1.66	1.84	2.01
0.75	0.46	0.67	0.83	0.95	1.05	1.22	1.37	1.49
1.00	0.33	0.50	0.62	0.70	0.78	0.92	1.03	1.14
1.50	0.20	0.30	0.38	0.45	0.50	0.60	0.67	0.74
2.00	0.14	0.20	0.27	0.32	0.35	0.42	0.47	0.51
2.50	0.09	0.14	0.19	0.22	0.24	0.28	0.32	0.35
3.00	0.07	0.11	0.14	0.17	0.18	0.21	0.23	0.26
3.50	0.06	0.09	0.11	0.13	0.14	0.16	0.19	0.21
4.00	0.05	0.07	0.09	0.10	0.12	0.14	0.15	0.17
5.00	0.03	0.05	0.07	0.08	0.09	0.10	0.12	0.14
6.00	0.03	0.04	0.05	0.06	0.07	0.08	0.10	0.11
7.00	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
8.00	0.02	0.03	0.04	0.04	0.05	0.06	0.07	0.07
9.00	0.02	0.02	0.03	0.04	0.04	0.05	0.06	0.06
10.00	0.01	0.02	0.03	0.03	0.04	0.04	0.05	0.05

Table S2. The median UHS at the New Delhi site at eight return periods.

Period (s)	Return Period							
	500 years (g)	1000 years (g)	1500 years (g)	2000 years (g)	2500 years (g)	3500 years (g)	4500 years (g)	5700 years (g)
0.00	0.12	0.17	0.20	0.23	0.24	0.28	0.30	0.32
0.02	0.13	0.17	0.20	0.23	0.25	0.28	0.30	0.32
0.05	0.16	0.22	0.26	0.29	0.31	0.35	0.38	0.41
0.10	0.23	0.32	0.37	0.41	0.45	0.51	0.55	0.60
0.20	0.27	0.37	0.43	0.48	0.53	0.59	0.64	0.70
0.50	0.24	0.33	0.40	0.46	0.50	0.58	0.63	0.68
0.75	0.21	0.28	0.33	0.38	0.41	0.46	0.51	0.56
1.00	0.16	0.22	0.27	0.30	0.33	0.38	0.41	0.45
1.50	0.11	0.15	0.18	0.21	0.23	0.26	0.28	0.31
2.00	0.07	0.10	0.13	0.14	0.16	0.18	0.20	0.21
2.50	0.05	0.07	0.09	0.10	0.11	0.13	0.14	0.16
3.00	0.04	0.06	0.07	0.08	0.09	0.10	0.11	0.13
3.50	0.03	0.05	0.06	0.07	0.07	0.09	0.09	0.10
4.00	0.03	0.04	0.05	0.06	0.06	0.07	0.08	0.09
5.00	0.02	0.03	0.04	0.04	0.05	0.05	0.06	0.07
6.00	0.02	0.03	0.03	0.04	0.04	0.05	0.05	0.06
7.00	0.01	0.02	0.03	0.03	0.04	0.04	0.05	0.05
8.00	0.01	0.02	0.02	0.03	0.03	0.04	0.04	0.05
9.00	0.01	0.02	0.02	0.02	0.03	0.03	0.04	0.04
10.00	0.01	0.01	0.02	0.02	0.02	0.03	0.03	0.04

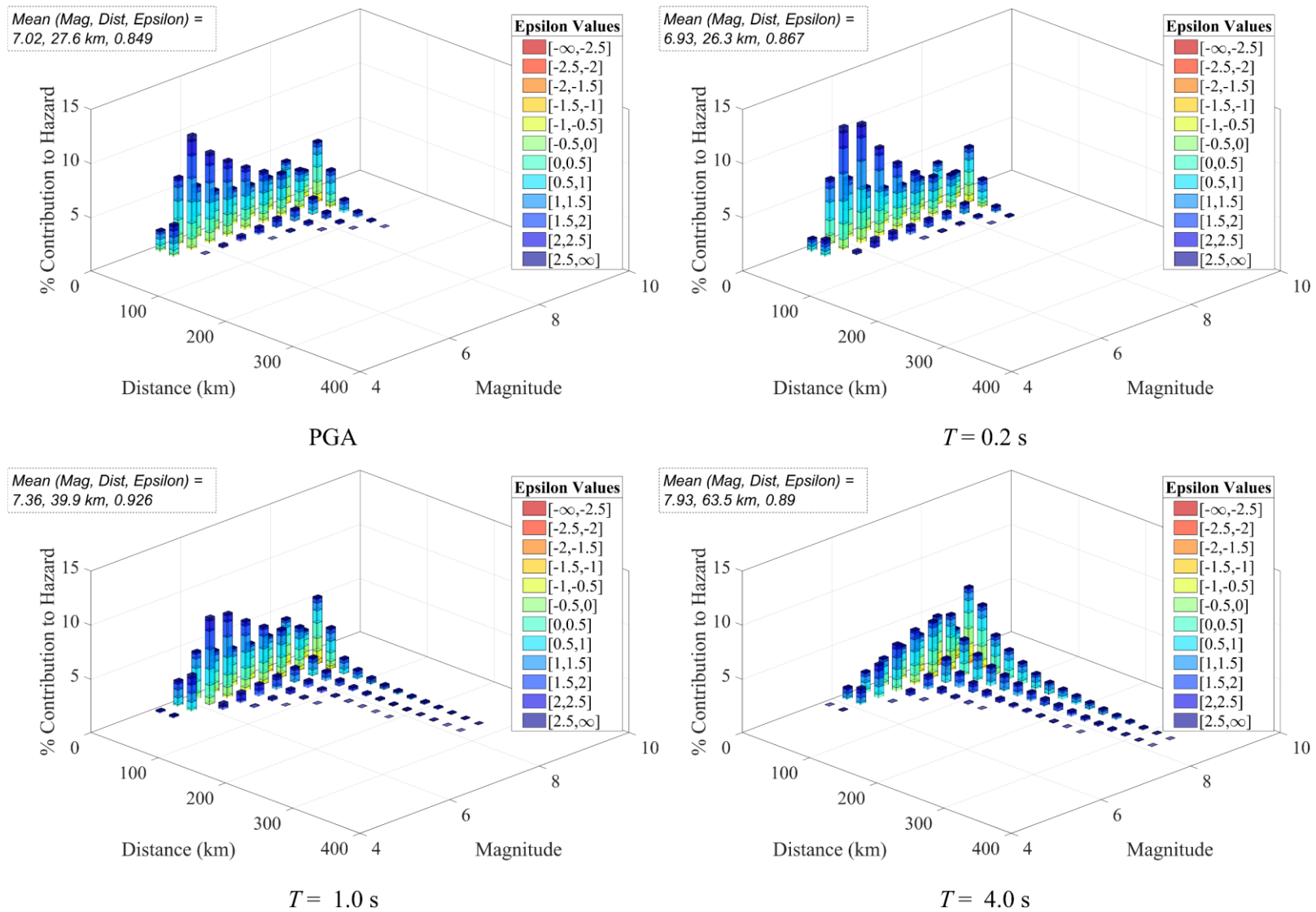


Fig. S1. Disaggregation of the hazard at the Dehradun site for 1000-year return period.

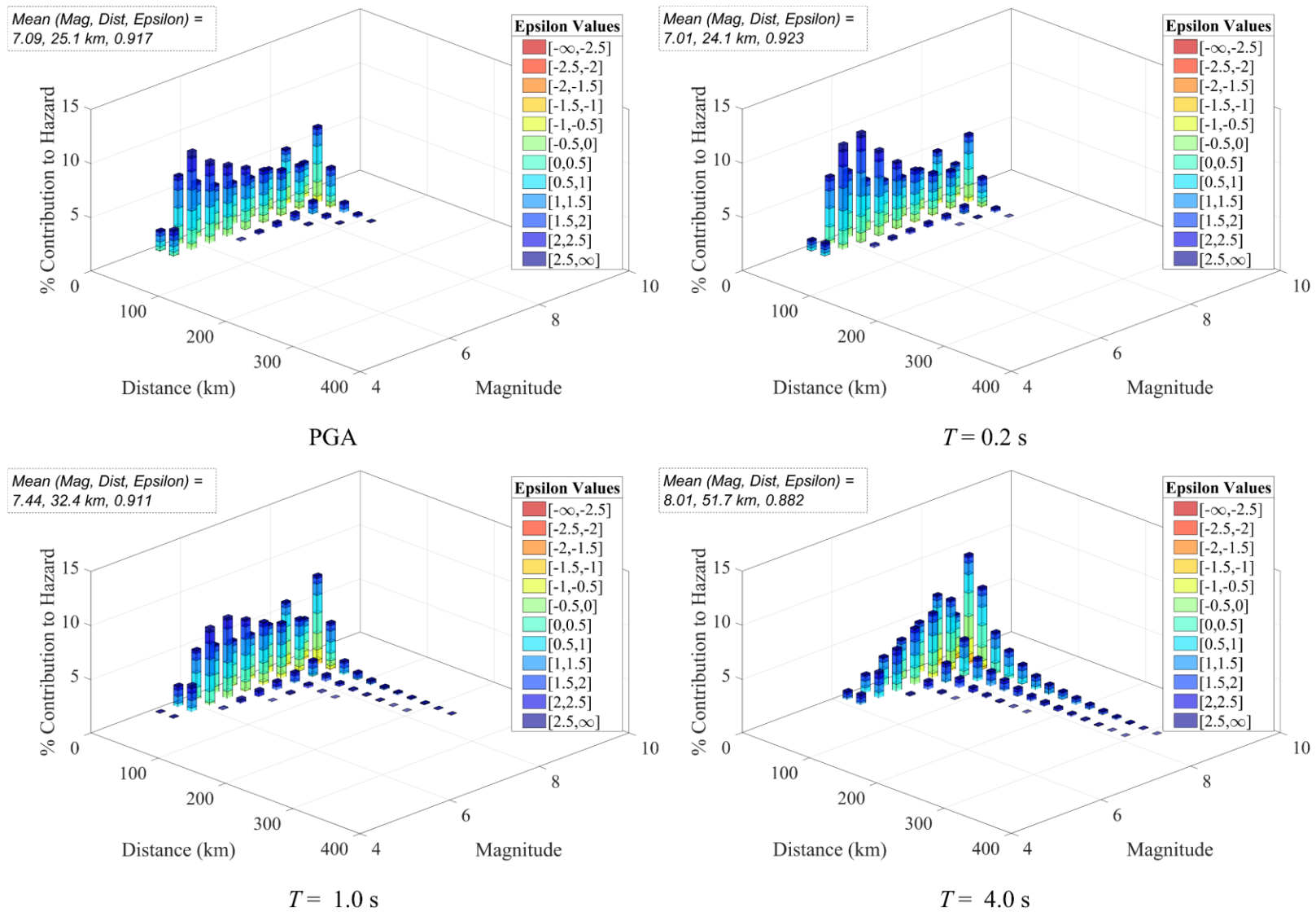


Fig. S2. Disaggregation of the hazard at the Dehradun site for 1500-year return period.

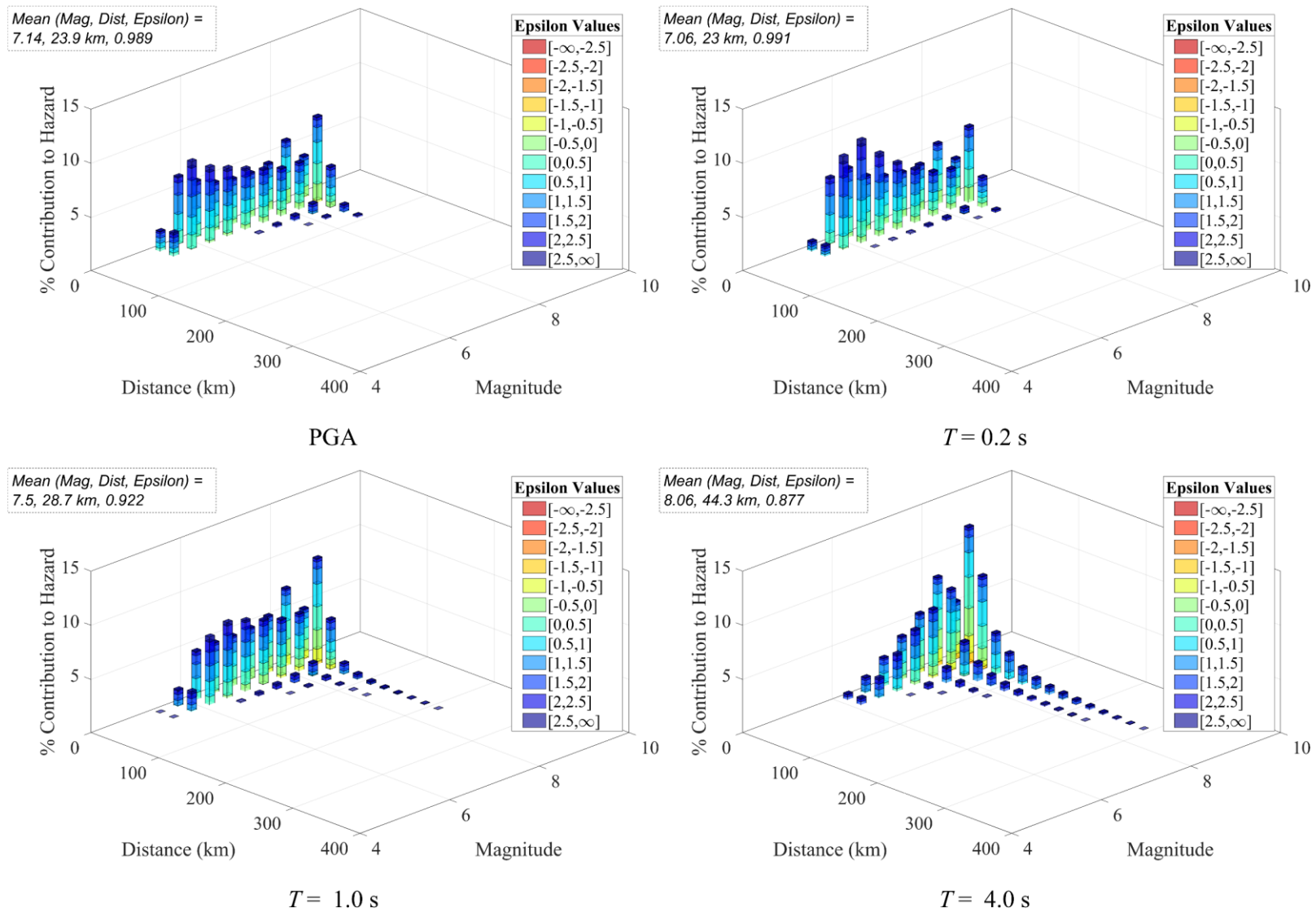


Fig. S3. Disaggregation of the hazard at the Dehradun site for 2000-year return period.

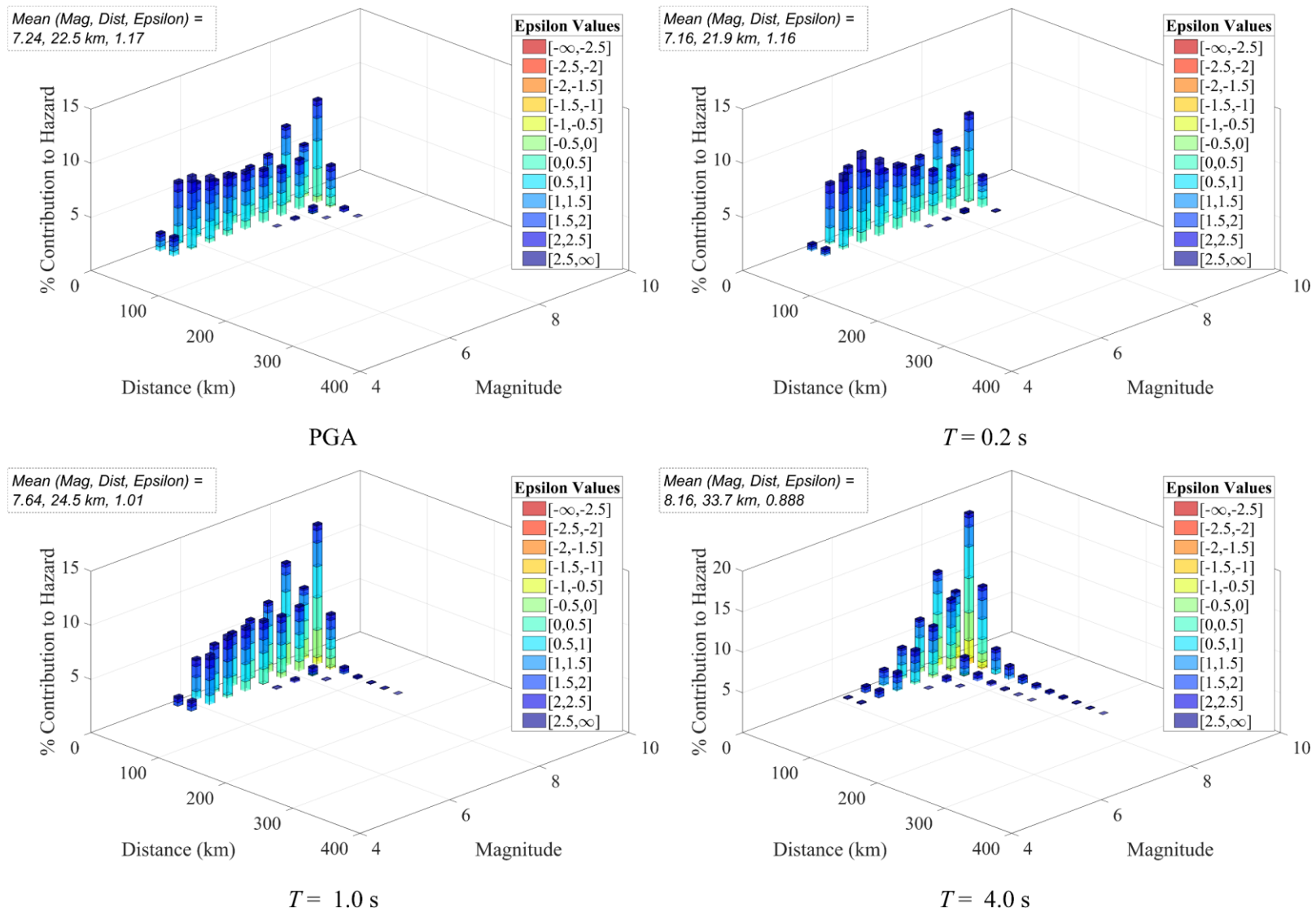


Fig. S4. Disaggregation of the hazard at the Dehradun site for 3500-year return period.

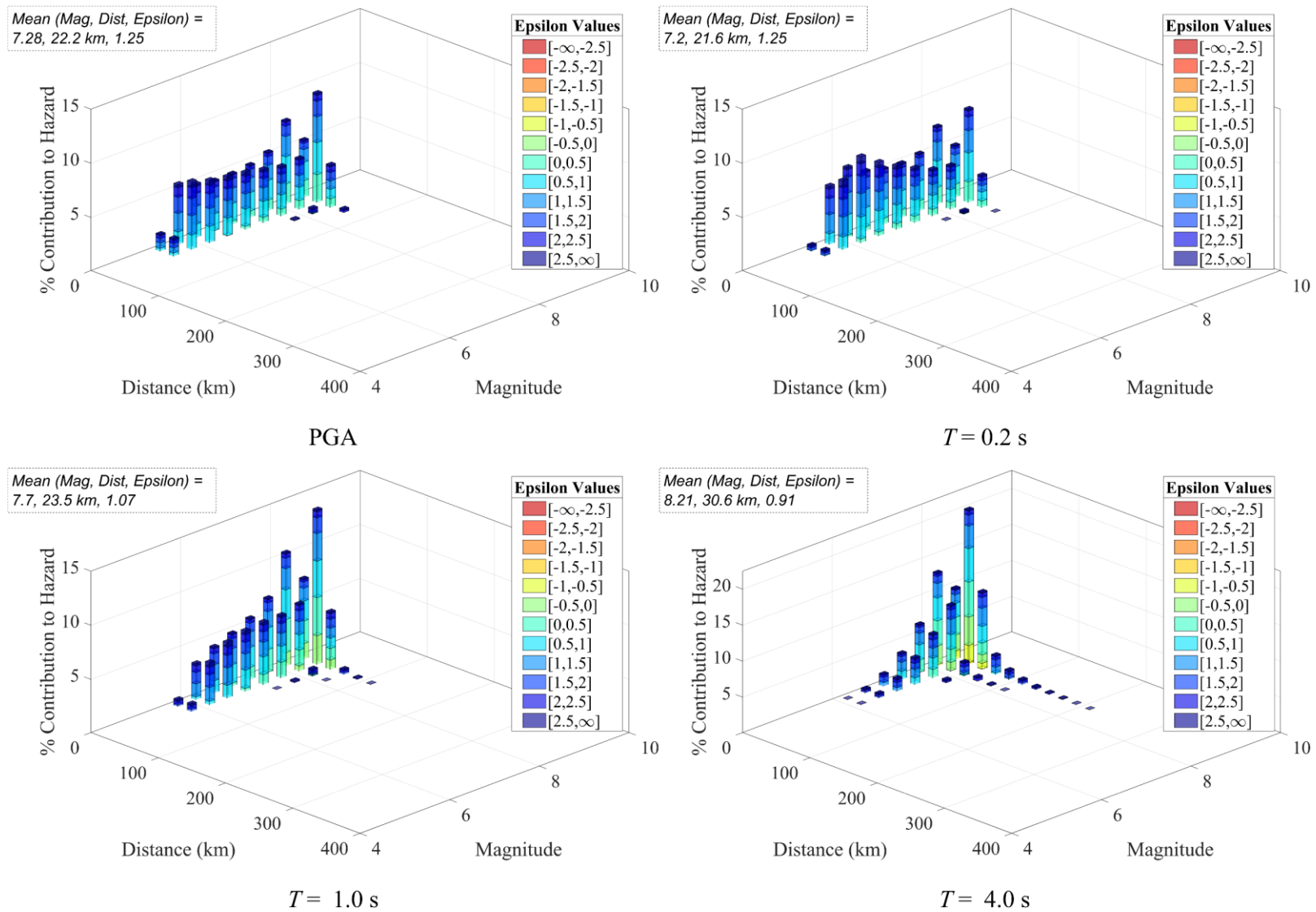


Fig. S5. Disaggregation of the hazard at the Dehradun site for 4500-year return period.

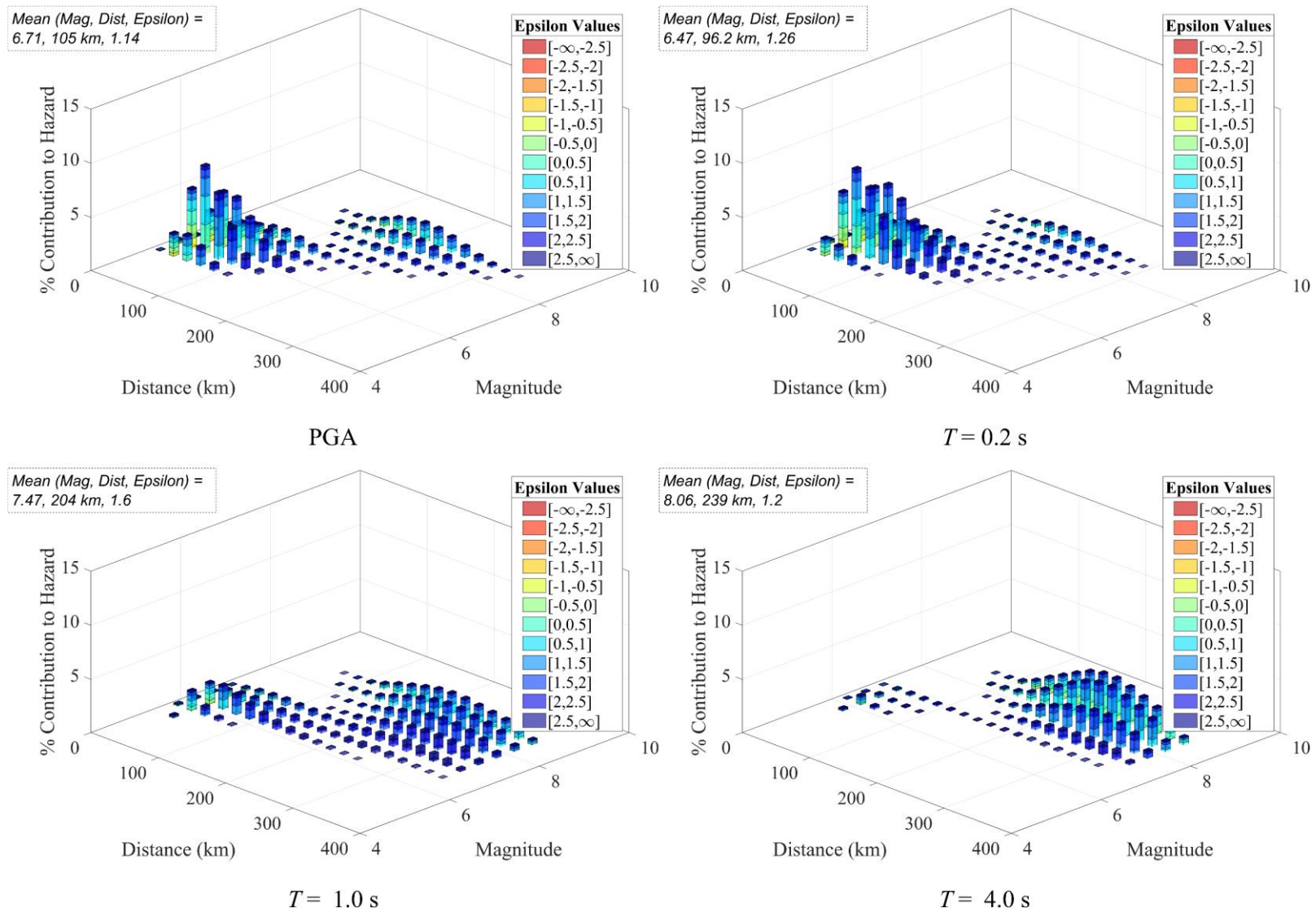


Fig. S6. Disaggregation of the hazard at the New Delhi site for 1000-year return period.

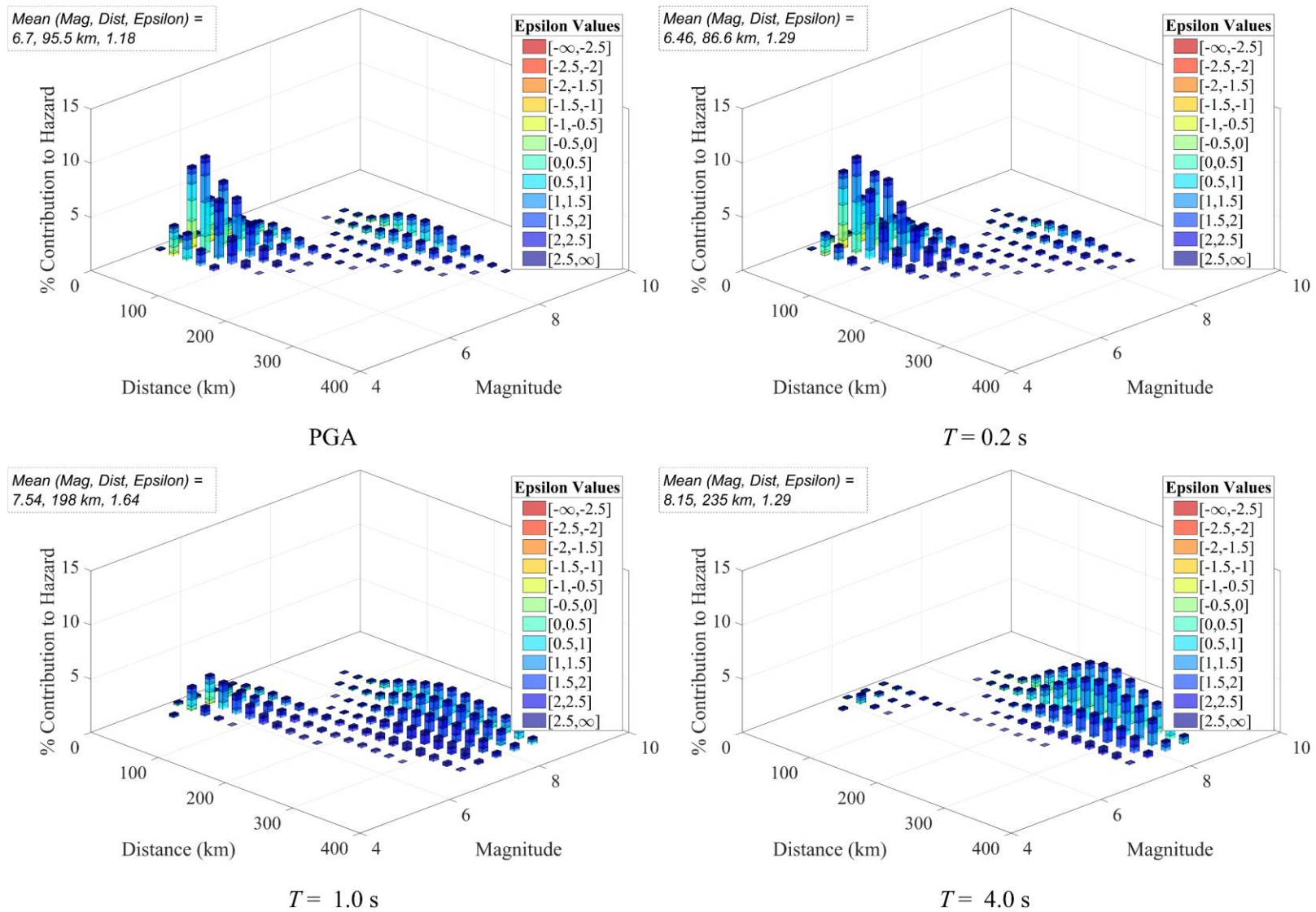
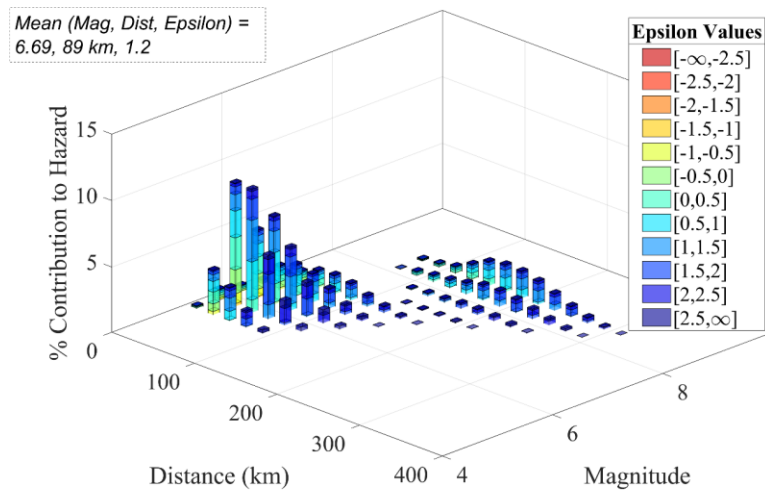
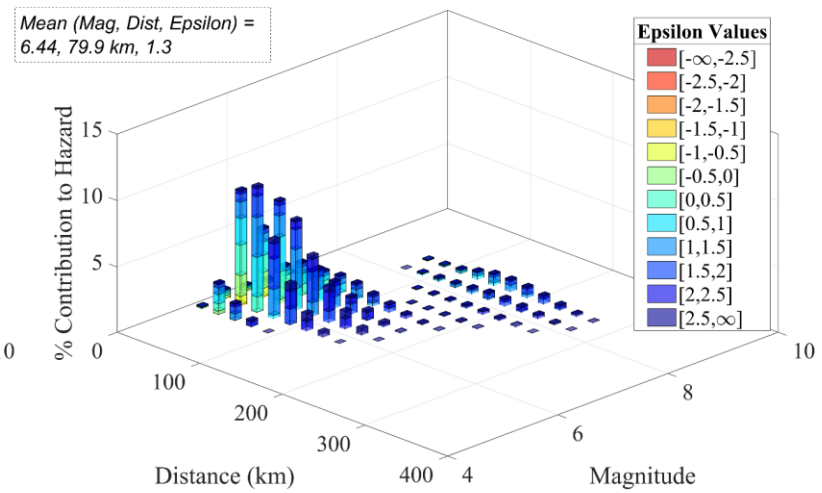


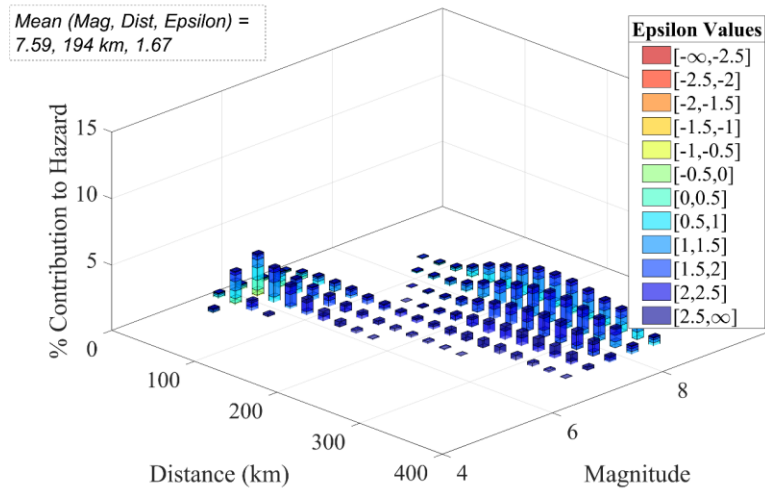
Fig. S7. Disaggregation of the hazard at the New Delhi site for 1500-year return period.



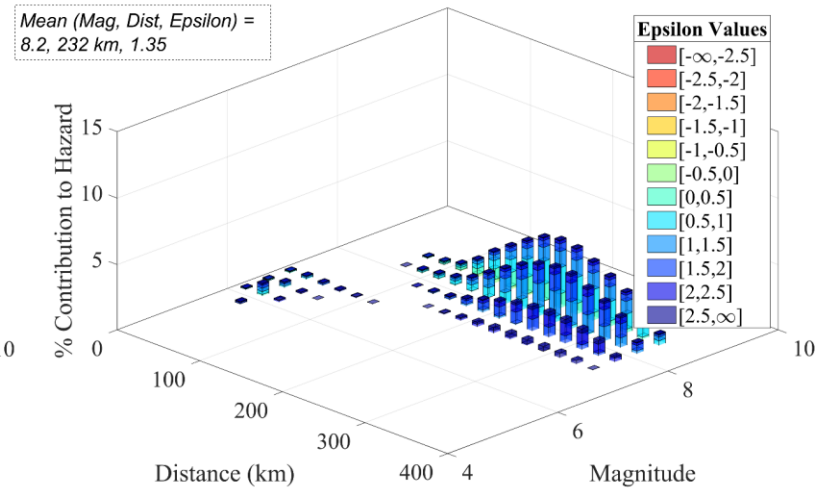
PGA



$T = 0.2$ s

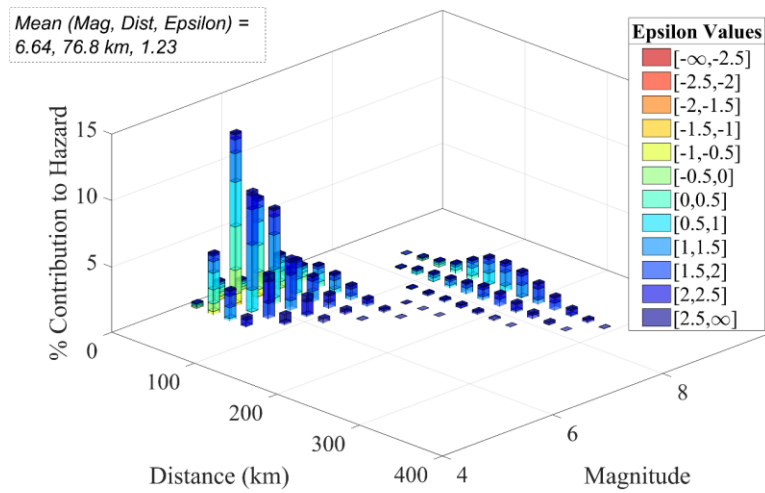


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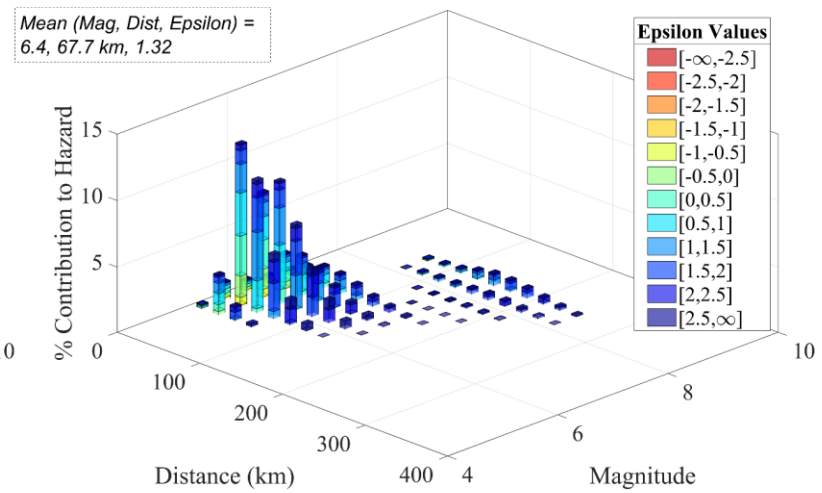


$T = 4.0$ s

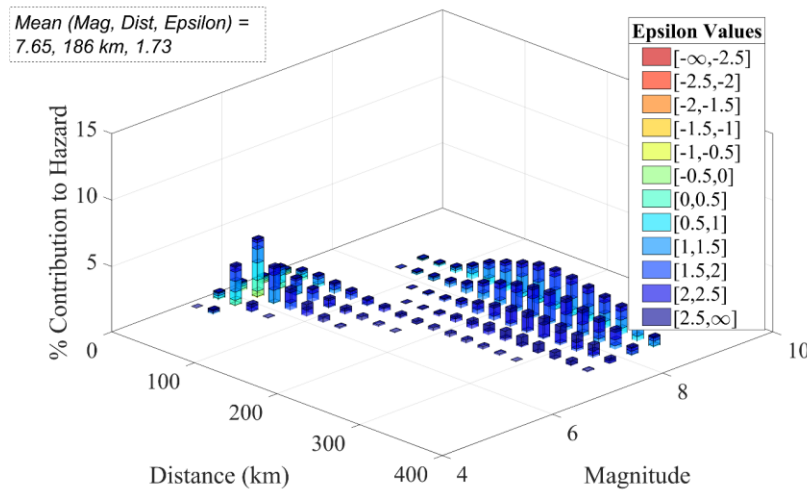
Fig. S8. Disaggregation of the hazard at the New Delhi site for 2000-year return period.



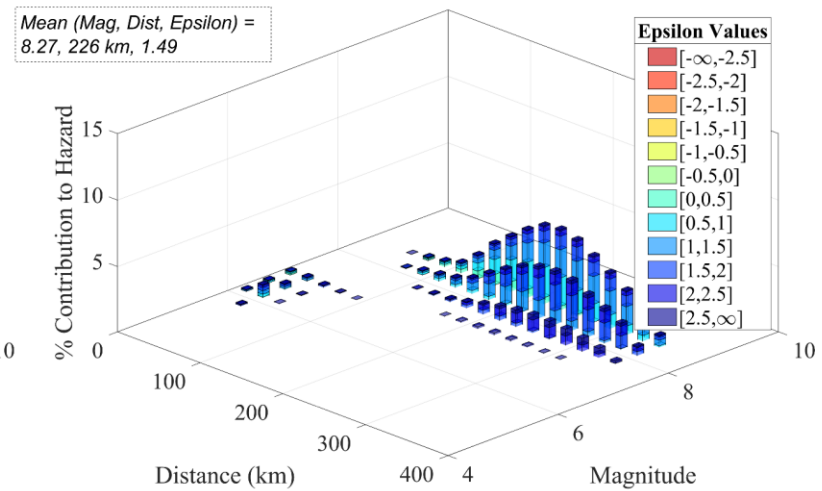
PGA



$T = 0.2$ s



$T = 1.0$ s



$T = 4.0$ s

Fig. S9. Disaggregation of the hazard at the New Delhi site for 3500-year return period.

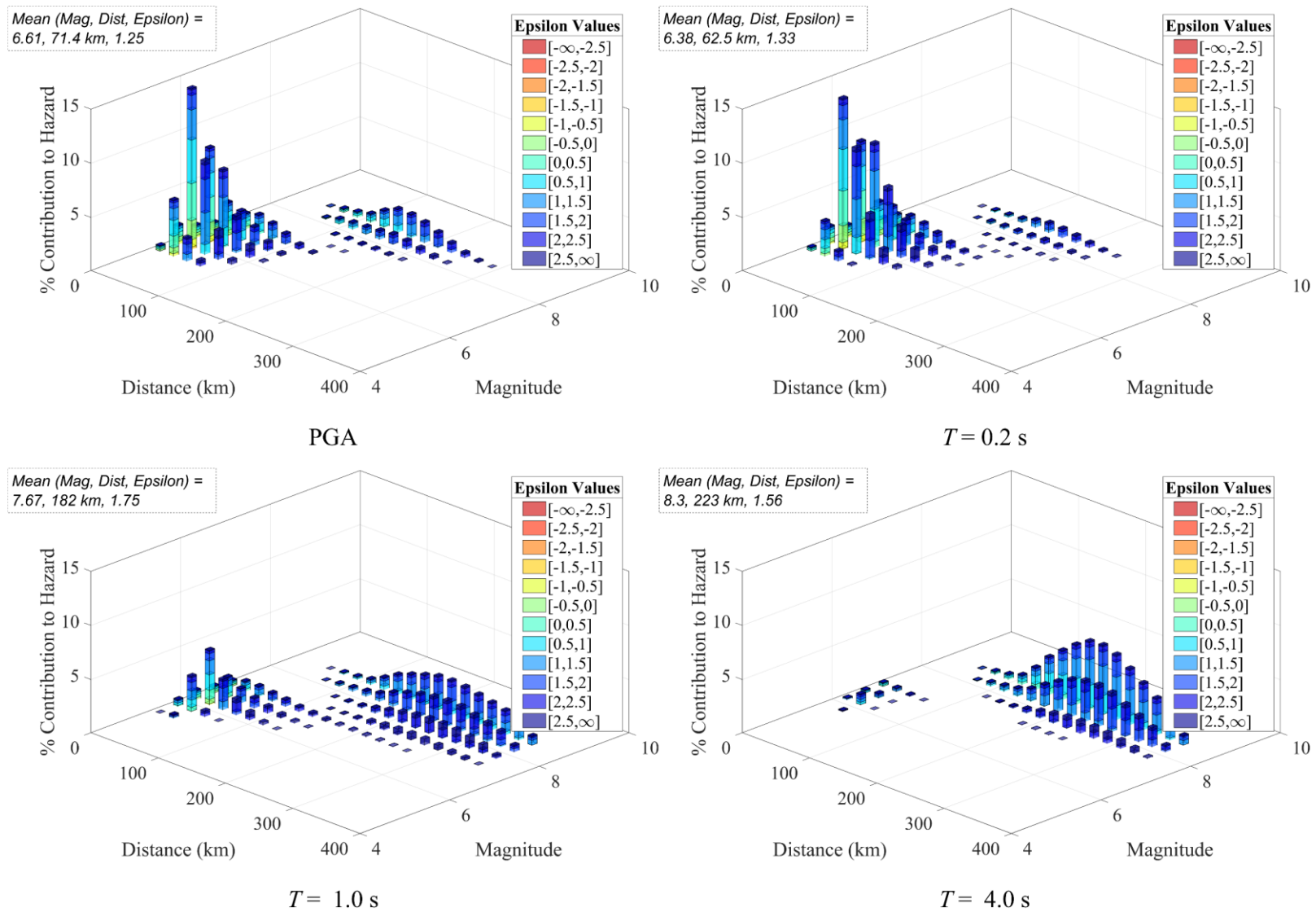


Fig. S10. Disaggregation of the hazard at the New Delhi site for 4500-year return period.